9.1 Changing employment structures

This chapter will explain:

★ how production is classified into different economic sectors
★ how employment structure varies and changes over time in LICs, MICs and HICs
★ the factors that affect the location and distribution of industries.

Classifying production into different economic sectors

In all modern economies people are employed across hundreds – and in some cases thousands – of different jobs. This wide range of jobs comprises four broad economic sectors:

» Primary
» Secondary
» Tertiary
» Quaternary.

The **primary sector** exploits raw materials. Farming (Figure 9.1), fishing, forestry and mining make up most jobs in this sector. Some primary products are sold directly to the consumer, but most go to secondary industries for processing. For example, fresh fish and fresh farm products are available to customers in supermarkets and other food retailers. However, large volumes of fish and farm products are sent to food processing factories to be frozen, tinned, packaged and so on. The drilling for crude oil is another example of a primary activity. However, this crude oil is of no use until it has been processed in an oil refinery into products such as petrol for motor vehicles and aviation fuel.

The **secondary sector** manufactures primary materials into finished products. Activities in this sector include the production of processed food (Figure 9.2), motor vehicles, clothing, furniture and building construction. Secondary products are classed as either consumer goods (produced for sale to the public) or capital goods (produced for sale to other industries).

The **tertiary sector** provides services to people and to businesses. Retail employees, delivery drivers, architects, nurses and teachers are examples of occupations in this sector. Transportation is a major tertiary industry that moves both people and freight over a wide range of distances. Figure 9.3 shows a freight train carrying sawn timber to be used directly in the construction industry or to be processed into furniture or other products. A modern economy could not function without a reasonably efficient transport sector.

▲ **Figure 9.1** The primary sector: a dairy farm, North Island, New Zealand

▲ **Figure 9.2** The secondary sector: a grain processing factory, Chicago, USA

▲ **Figure 9.3** The tertiary sector: a freight train on the Trans-Siberian Railway, carrying sawn timber from eastern Russia to Moscow

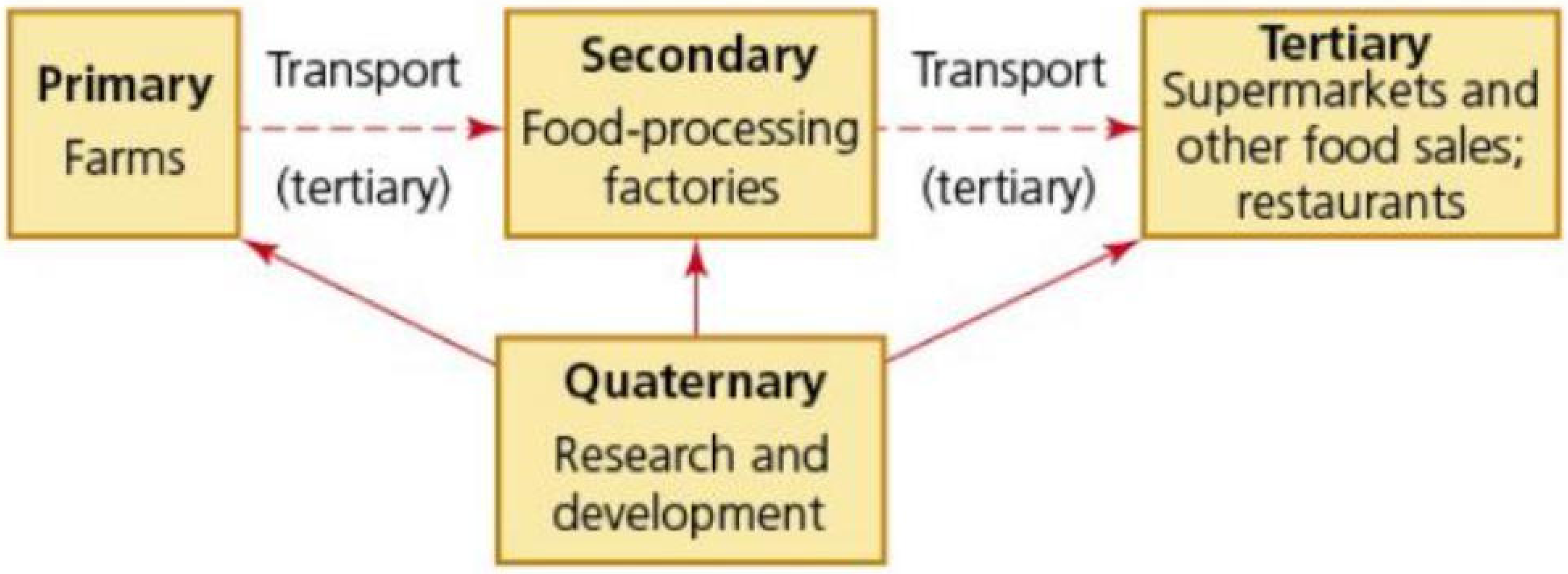
Banking is another important industry within the tertiary sector. Figure 9.4 shows a branch of the National Bank of Kenya in Nairobi's central business district, where many banks, both Kenyan and international, have branches.

▲ **Figure 9.4** The tertiary sector: the National Bank of Kenya in Nairobi, Kenya

The **quaternary sector** uses high technology to provide information and expertise to all sectors of an economy. It is sometimes referred to as the ‘knowledge-based part of the economy’. Jobs in this sector include aerospace engineers, research scientists (Figure 9.5), biotechnology workers and computer programmers. A well-functioning quaternary sector requires a highly educated workforce. The quaternary sector has only been recognised as a separate economic sector since the late 1960s. Before then jobs that are now classed as quaternary were placed in either the secondary or tertiary sectors, depending on whether or not they produced a tangible product. However, even today much of the available information on employment does not consider the quaternary sector.

▲ **Figure 9.5** The quaternary sector: a research scientist

A **product chain** can be used to show the relationship between the four sectors of employment. It is the full sequence of activities needed to turn raw materials into a finished product. The food industry (Figure 9.6) provides a good example. Some large companies are involved in all four stages of the food product chain.



▲ **Figure 9.6** The food industry’s product chain

Activities

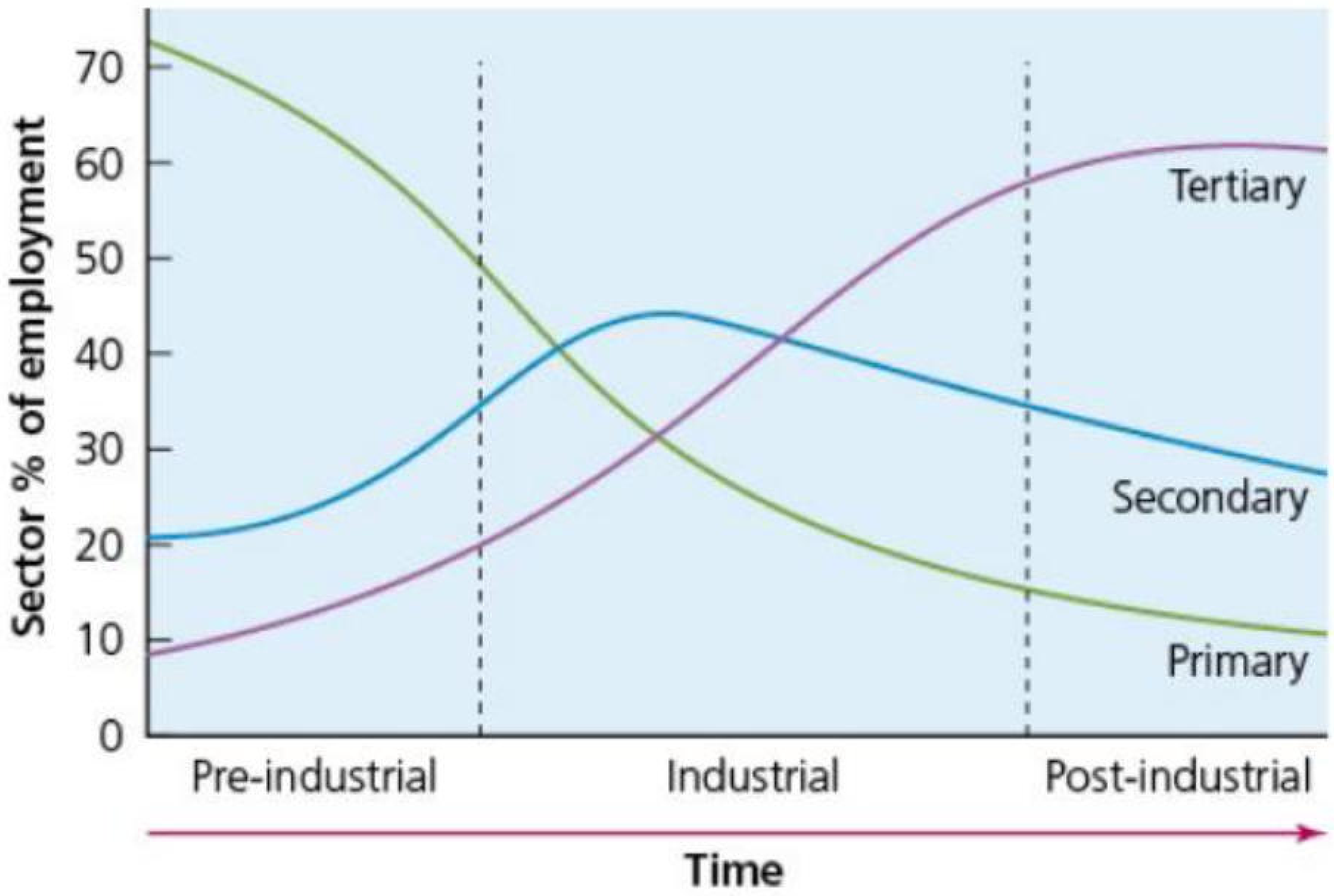
1 Define the terms:
 a Primary sector
 b Secondary sector
 c Tertiary sector
 d Quaternary sector.
2 Give two examples of jobs in each of the four economic sectors.
3 Describe the food industry product chain shown in Figure 9.6.
4 What job do you want to do when you complete your education? In which sector of employment is this job?

LICs, MICs and HICs – how employment structure varies and changes over time

As an economy develops, the proportion of people employed in each economic sector changes. The Clark-Fisher economic sector model illustrates these changes (Figure 9.7). The model covers three economic stages:

- Pre-industrial (LICs)
- Industrial (MICs)
- Post-industrial (HICs).

In the process of economic development, countries move from one stage to another. Such movement occurs at different times and speeds, depending upon economic conditions in individual countries. Critics of the model question its linear view of development, which was based on the experience of Western economies. They argue that a country's level of development cannot be solely based on its major industries.



▲ **Figure 9.7** The Clark-Fisher economic sector model

LICs – pre-industrial

This is when the economy of a country is based mainly on primary activities. In this stage, most people work in agriculture and other primary industries, which are very **labour intensive**. These are largely rural economies, dominated by **subsistence agriculture**. The secondary and tertiary sectors are very limited in scope, and are also labour intensive. Many countries in this stage have faced difficulties in developing their secondary and tertiary sectors, due to historical and ongoing challenges such as poor access to markets of HICs and limited representation within important international organisations. Many countries are still at this stage, such as are Papua New Guinea (Figure 9.8), Nepal and Afghanistan.

▲ **Figure 9.8** Papua New Guinea: a pre-industrial economy – here small farmers are selling produce in a local market in the Western Highlands

MICs – industrial

This stage begins when an individual country starts its '**Industrial Revolution**'. The UK was the first country in the world to undergo such a transformation, which began in the late eighteenth century. This was due to a combination of technological development in the UK and low-cost imports of important commodities from locations such as India and the Caribbean, over which the UK held colonial control. Some economists such as A.G. Frank have argued that colonisation held back social and economic development in many countries that were colonies.

Other leading nations such as Germany and the USA followed the industrialisation of the UK within a few decades. The developed countries in the world today began to exit the industrial economic stage from the 1960s. However, the 1960s was also the time when an increasing number of developing countries became industrialised, and were considered '**newly industrialised countries**' (emerging economies).

» Four countries stood out as the first generation of newly industrialised countries – South Korea, Taiwan, Hong Kong and Singapore (Figure 9.9).
» Such was the rapid rate of economic growth in these countries, that they were referred to as the 'Asian

tigers'. A tiger economy is one that is experiencing very rapid growth.

- The emergence of newly industrialised countries is often associated with the changing nature of their relationships with Western economies in the post-colonial era.
- Within a few decades, a second generation of newly industrialised countries emerged, including Malaysia, Indonesia and Thailand.
- Within an even shorter time interval (10 to 20 years), the economies of a third group of nations began to expand rapidly, led by China and India.
- In emerging economies, production and employment in manufacturing has increased rapidly in the last 30 to 60 years.

▲ **Figure 9.9** The central business district, Singapore

HICs – post-industrial

The HICs of today do not include just the traditional developed regions of the world, but also the first generation of newly industrialised countries. This is because their industries have moved up the 'technology ladder', and their tertiary and quaternary sectors have expanded rapidly.

Countries such as the USA, the UK, Japan (Figure 9.10) and Germany are **post-industrial societies**, with most people employed in the tertiary sector. The UK exemplifies the way in which most HICs have experienced all three stages:

▲ **Figure 9.10** Japan's high-speed Bullet Train, in Tokyo Central Railway Station

- In 1800, about 75 per cent of employment in the UK was in the primary sector, with about 15 per cent in the secondary sector and 10 per cent in the tertiary sector.
- The Industrial Revolution resulted in massive changes in the UK economy. By 1900, the respective figures in the UK's economic sectors were 30 per cent primary, 55 per cent secondary and 15 per cent tertiary.
- As the twentieth century progressed, technological advance resulted in increasing **mechanisation** in primary industries, reducing the demand for workers in this sector. By 2022, only 1.2 per cent of UK workers were employed in the primary sector.
- While the share of workers employed in the secondary sector varied somewhat in the early part of the twentieth century, the period of most significant and continuous decline did not begin until the mid-1960s. Unsurprisingly, this was also the beginning of a rapid increase in the percentage employed in the tertiary sector.
- The decline in manufacturing employment was due to a range of factors, the most important of which were **deindustrialisation**, globalisation, automation and government policies. In terms of automation,

most manufacturing plants became more **capital intensive**.

- » The tertiary sector has also changed as computer networks and other technical advances such as artificial intelligence have reduced the number of people required in some occupations. But, for some service industries such as healthcare and tourism, employment has increased in many countries.
- » Employment in the quaternary sector has become more important, which can be seen as a good measure of how advanced an economy is.

Table 9.1 compares the employment structure of an HIC, an MIC and an LIC. The contrasts are considerable, particularly between the contributions of the primary and tertiary sectors, and are a reasonable fit with the sector model presented in Figure 9.7 (page 210).

A comparison of Tables 9.1 and 9.2 shows that there is a very clear link between employment structure and socioeconomic development.

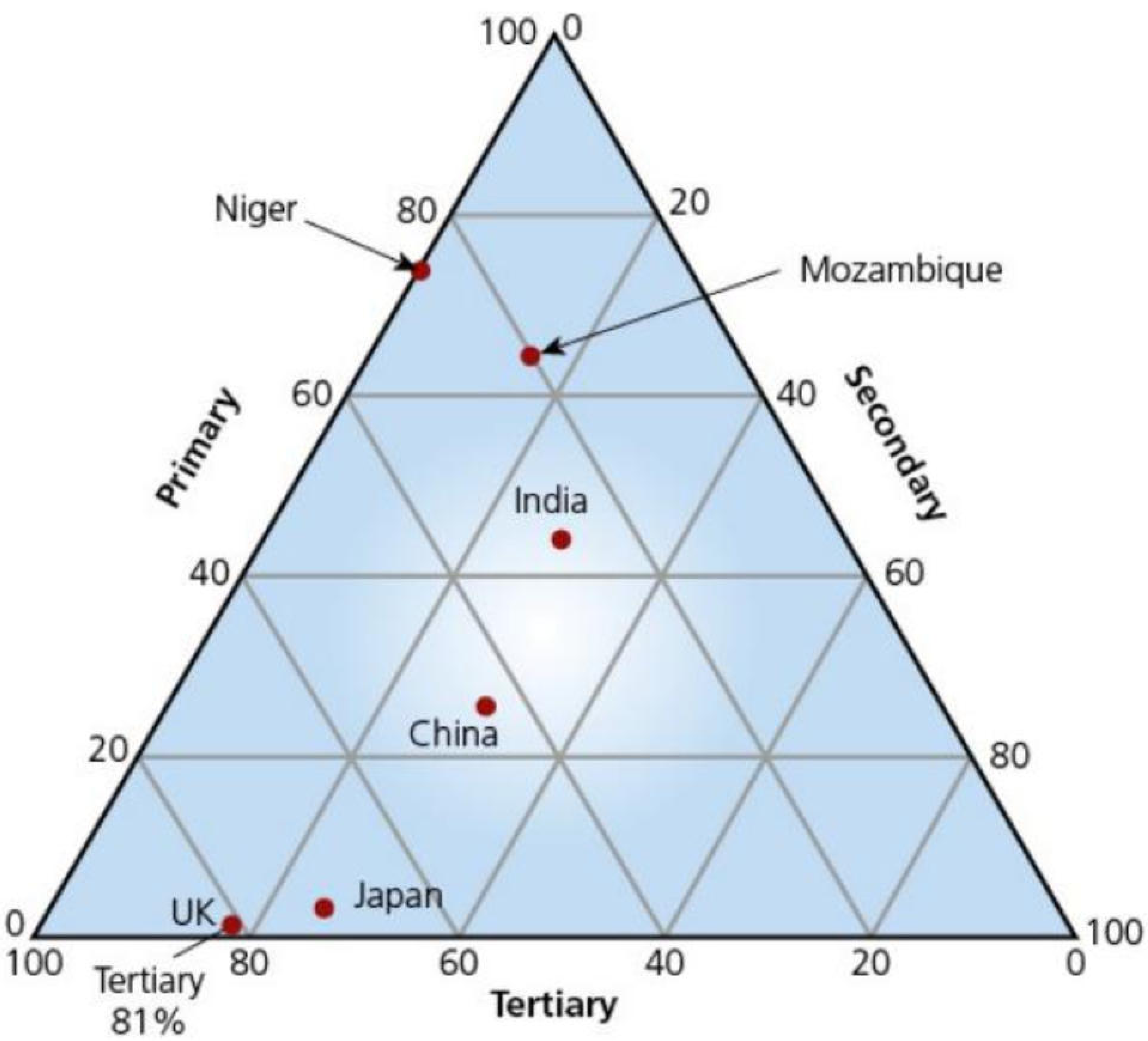
▼ **Table 9.1** The employment structure of an HIC, an MIC and an LIC (data for 2022)

Country	% primary	% secondary	% tertiary
France	2.6	19.3	78.2
Brazil	8.7	20.5	70.8
Bangladesh	36.9	21.9	41.4

▼ **Table 9.2** Development indicators for France, Brazil and Bangladesh

Country	GNI per capita (PPP)	Infant mortality rate	Life expectancy at birth
France	57,090	3.2	82
Brazil	17,260	11	73
Bangladesh	7690	25	74

A graphical method often used to compare the employment structure of countries is the **triangular graph** (Figure 9.11).



▲ **Figure 9.11** Triangular graph

- » One side (axis) of the triangle is used to show the data for each of the primary, secondary and tertiary sectors.
- » Each axis is scaled from 0 to 100 per cent.
- » The indicators on the graph show how the data for the UK can be read.

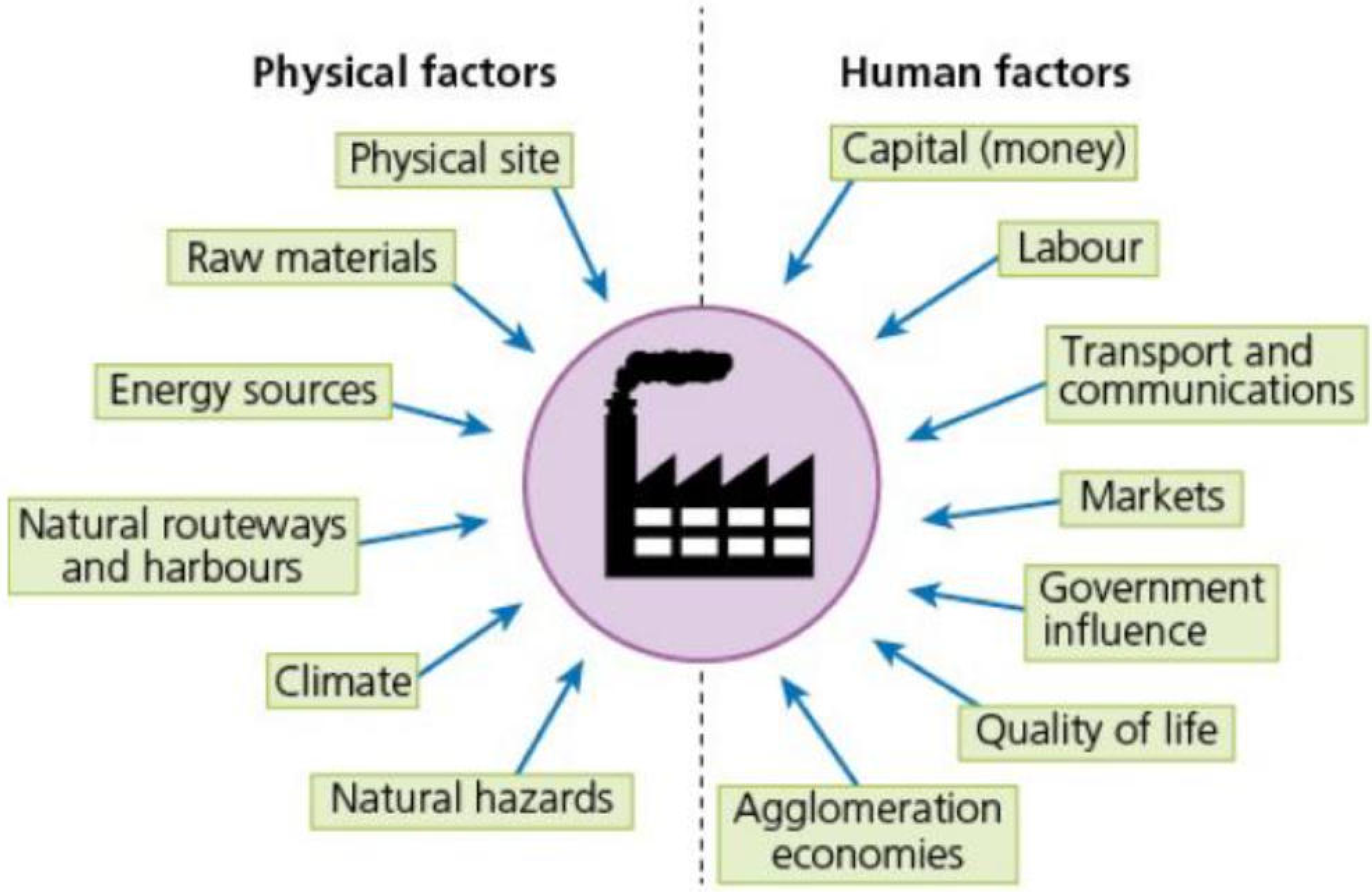
Figure 9.11 shows data for two HICs, two MICs and two LICs.

Activities

1 Look at Figure 9.7 (page 210). Describe and explain how the employment structure of developed countries has changed over time.
2 Define the terms:
   a Post-industrial society
   b Deindustrialisation.
3 Look at Table 9.1. Plot the employment data for France, Brazil and Bangladesh on a copy of the triangular graph shown in Figure 9.11.

The factors affecting the location and distribution of industries

Every day, decisions are made about where to locate industrial premises, ranging from small workshops to huge industrial complexes. In general, the larger the company, the greater the number of real alternative locations available. For each possible location, a wide range of factors can have an impact on total costs and thus influence the decision-making process. The factors affecting industrial location differ from industry to industry and their relative importance is subject to change over time. These factors can be broadly subdivided into physical and human (Figure 9.12). They relate both to individual factories and to industrial zones.



▲ **Figure 9.12** The physical and human factors influencing industrial location

The physical factors influencing industrial location

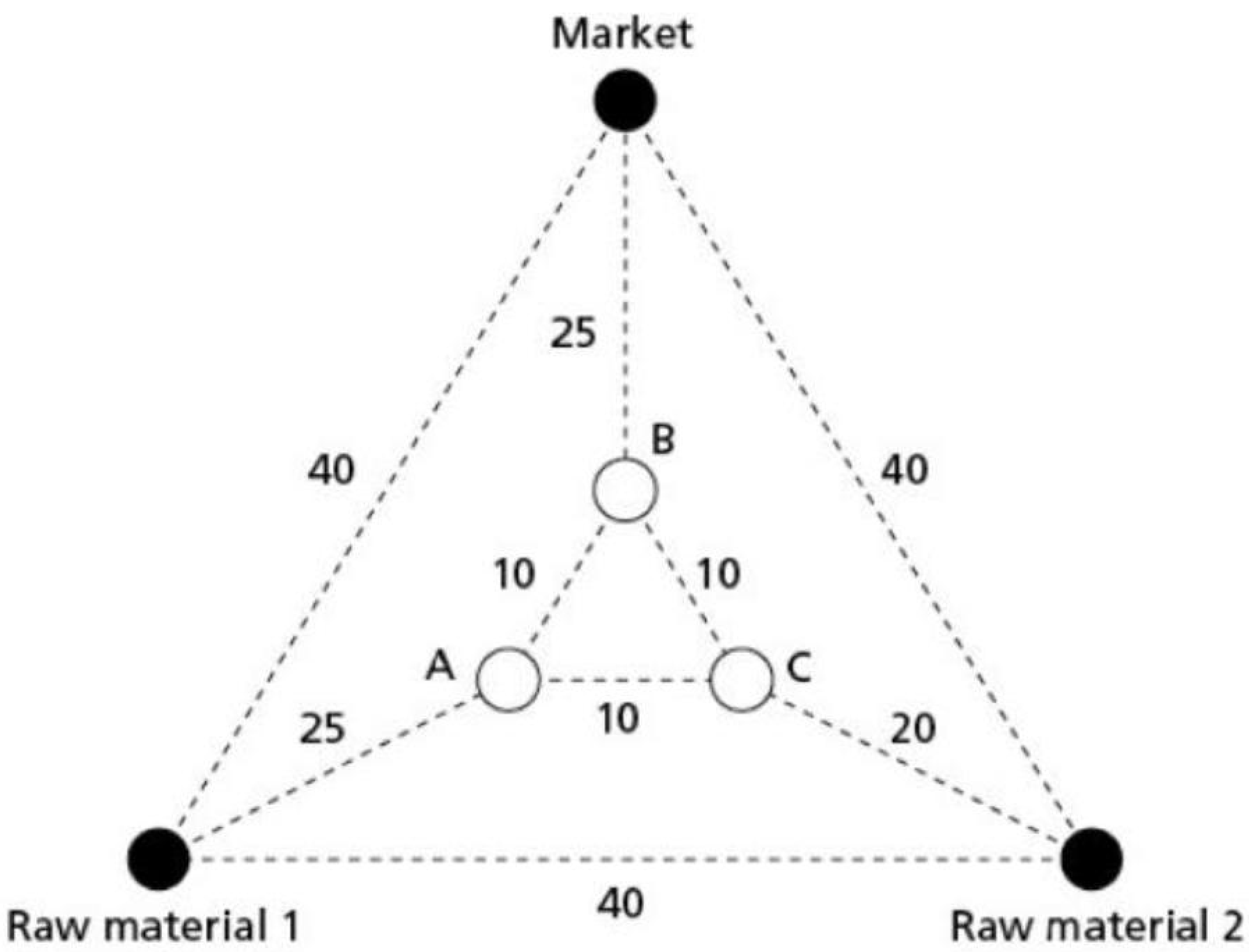
Physical site

The availability and cost of the land area required for the site of a factory or other business is a fundamental factor in location. Large factories need flat, well-drained land on solid bedrock. An adjacent water supply may be essential if very large quantities of water are required for industrial production. The space requirements can vary enormously for different industries. Technological advance has made modern industry much more space-efficient than in the past. However, modern industry is more horizontally structured (on one floor) as opposed to, for example, the textile mills of the nineteenth century, which were on four or five floors. For modern developments such as **industrial estates**, retail parks and leisure centres, a large site and a highly accessible location are important.

In the UK during the Industrial Revolution, entrepreneurs faced relatively few legal restrictions on location compared with today. However, over time more and more areas have been placed off-limits to industry, mainly to conserve the environment. Areas such as National Parks, Country Parks and Areas of Special Scientific Interest now occupy a significant portion of the UK. In urban areas, land-use zoning restricts location and **green belts** often prohibit location at the edge of urban areas.

Raw materials

Industries requiring heavy and bulky raw materials tend to locate as close as possible to the raw material deposits. This is mainly to minimise transport costs. The processes involved in turning raw materials into a manufactured product usually result in **weight loss**. This means that the weight/bulk of the finished product leaving the factory is much less than the weight/bulk of the raw materials needed to produce it. Therefore, the transport costs of bringing the raw materials to the factory will be greater than the cost of moving the finished product to market (Figure 9.13). The clearest examples of this influence are where only one raw material is used. Figure 9.14 shows a very large, modern pulp and paper mill in British Columbia, Canada. It is located within an enormous forested area that supplies the mill with wood. In the UK, sugar beet refineries are centrally located in crop-growing areas because there is a 90 per cent weight loss in manufacture.



▲ **Figure 9.13** An example of weight loss in manufacturing

▲ **Figure 9.14** A pulp and paper mill, British Columbia, Canada, located within the forests that provide its raw material

Large and well-equipped sea ports are good locations for industries that use significant quantities of imported raw materials. Examples include flour milling, food processing, chemicals and oil refining. Such locations are **break-of-bulk points**, where cargo is unloaded from bulk carriers and transferred to smaller units of transport for onward transportation. If raw materials are processed at the break-of-bulk point, it can result in significant savings in transport costs.

Activity

Look at Figure 9.13 (page 213), which illustrates the concept of weight loss. The triangle shows the location of:

- the two raw materials used to manufacture a product
- the single market in which the product is sold
- the road network and the distance in kilometres between road junctions
- the three possible locations for the factory, A, B or C.

Two tonnes of raw material 1 and one tonne of raw material 2 are required to manufacture one tonne of the finished product. Transport costs are £1 per tonne per km. Construct a table to calculate which of the three potential factory locations, A, B or C, is the lowest transport-cost location.

Energy sources

The Industrial Revolution in the UK and many other countries was based on the use of coal as a fuel. The coal was often much more costly to transport than the raw materials required for processing. It is therefore not surprising that outside of London, most of the UK's industrial towns and cities developed on coalfields or ports close to coalfields. However, during the twentieth century the construction of national electricity grids and gas pipeline systems made energy a virtually ubiquitous resource in HICs. However, even in HICs there are some industries that are constrained in terms of location because of a very high energy requirement. For example, the lure of low-cost hydroelectric power has resulted in a large concentration of electro-metallurgical and electrochemical industries in southern Norway.

Natural routeways and harbours

Many modern roads and railways follow natural routeways. A good example is the gaps eroded by rivers in chalk escarpments. Settlements with accompanying economic activity are typically located at both the scarp slope (the steep slope of an escarpment) and the dip slope (the more gentle slope of an escarpment) entry points of these natural routeways.

A natural harbour is a landform where part of a body of water is protected from storm conditions and is deep enough to provide an anchorage. Sydney Harbour (Figure 9.15), Australia is generally recognised to be the largest and deepest natural harbour in the world.

▲ **Figure 9.15** Sydney, Australia

Climate

Some industries, such as the aerospace and film industries, benefit directly from a sunny climate. For example, a climate that has a large number of cloud-free days provides the aerospace industry with more opportunities to carry out test flights. It is no coincidence that the Hollywood film industry is in southern California. Indirect benefits include lower heating bills and a more favourable quality of life. Locations with extreme climates and those regions experiencing climatic hazards usually struggle to attract business investment unless there is a very compelling reason to locate in such an environment.

Natural hazards

Regions that experience devastating natural hazards are likely to lose out on investment to other regions where severe natural hazards do not occur. Apart from loss of life, infrastructure, workplaces and vital public services can be destroyed and transport links severed. When such disasters occur, companies may decide to rebuild in an environment with less risk. Natural hazards and the ability to deal with the consequences of natural hazards can limit economic development in LICs.

Activities

1 Give two examples of locations where companies would be unlikely to be granted planning permission to build a factory or retail park.
2 What are the advantages for some types of manufacturing industry in locating at or close to a large international port?
3 Give two examples of industries where climate might be an important location factor.
4 Why do countries subject to serious natural hazards often find it hard to attract foreign direct investment?

The human factors influencing industrial location

Capital (money)

Some areas are more likely to attract business investment than others. Countries and regions within countries with a reputation for business success are much more likely to attract investment in the secondary and tertiary sectors compared to locations perceived to be less successful. For example, Indonesia is an emerging economy that has attracted a high level of **foreign direct investment** (Figure 9.16). This is due to its large population of over 275 million, which is a huge internal market, along with an attractive business environment encouraged by government, abundant natural resources and a skilled population.

It is often more difficult to attract investment in LICs than in HICs and MICs because of the perceived risk. Political unrest and armed conflict deter investment.

Labour

The quality and cost of labour are important location factors (Figure 9.17). Although all industries have become more capital intensive over time, labour still accounts for about 20 per cent of total costs in manufacturing industries.

▲ **Figure 9.16** A large capital input was required to build this container port in Indonesia

▲ **Figure 9.17** Tapestry making is labour-intensive production in Hanoi, Vietnam

Variations in labour costs can be identified at different scales. Table 9.3 provides data for eight of the 27 countries within the European Union to show how much hourly labour costs varied in 2022. In general, labour costs within the EU are higher in Northern and Western Europe than they are in Eastern and Southern Europe. By far the greatest disparity is at the global scale. One of the major reasons transnational corporations have been attracted to emerging and developing economies since the 1960s has been the relatively low labour costs in these economies. The reputation, turnover, mobility and quantity of labour are other relevant factors.

▼ **Table 9.3** Variations in labour costs within the European Union, 2022

Country	**Hourly labour costs (US$)**
Luxembourg	53.39

France	42.96
Germany	41.60
Italy	30.96
Portugal	16.95
Poland	13.17
Hungary	11.26
Bulgaria	8.60

Transport and communications

Although once a major locational factor, the share of industry's total costs accounted for by transportation has fallen steadily over time. However, the quality of transport links and their subsequent reliability remains an important issue in locational decision-making. Transport costs are particularly important for heavy and bulky goods. **Accessibility** to airports, ports, motorways and so on may be crucial for some industries, such as those producing perishable goods.

Markets

Where a company sells its products may be a big influence on where a factory is located. Not all production processes result in weight loss. Industries that add substantial amounts of water to the other raw materials required, such as soft drinks and brewing, are examples of **weight-gain** industries. These need to be located close to their markets.

However, there are other reasons for market location. Industries where fashion and taste are variable need to be able to react quickly to changes demanded by customers. One of the reasons that the major vehicle manufacturing companies are spread around the world is to respond to customers' demands, which can vary substantially in different parts of the world.

Government influence

Government policies and decisions can have a large direct and indirect impact on the location of industry. In centrally planned communist countries, such as China and Cuba, the influence of government on industry is very strong. In other countries, for instance in Northern America or the European Union, the significance of government intervention depends on:

- the degree of public ownership of an economy – for example, in some countries industries such as water, energy and railways are run by public corporations, while in other countries they are run by independent companies responsible to their shareholders
- the strength of regional policy in terms of restrictions and incentives.

Governments influence industrial location for economic, social and political reasons. Government at national and regional levels may use grants to attract major international companies. There is a high level of competition both between and within countries to attract inward investment. Significant improvements in infrastructure may be important in attracting companies to locate in a country or region. Examples of major government spending policies include building new high-speed railways and airport expansion. Such decisions can affect levels of employment in different sectors of an economy.

National governments may also establish special economic zones, such as **freeports**, in which imported goods can be held or processed free of customs duties before re-export. Figure 9.18 illustrates the multiplier effect that a successful freeport can have on a regional economy. There are many examples of freeports around the world, including Shanghai (China), Barcelona (Spain), Cork (Ireland), Teesside (UK), Manaus (Brazil) and Bangkok (Thailand).

Quality of life

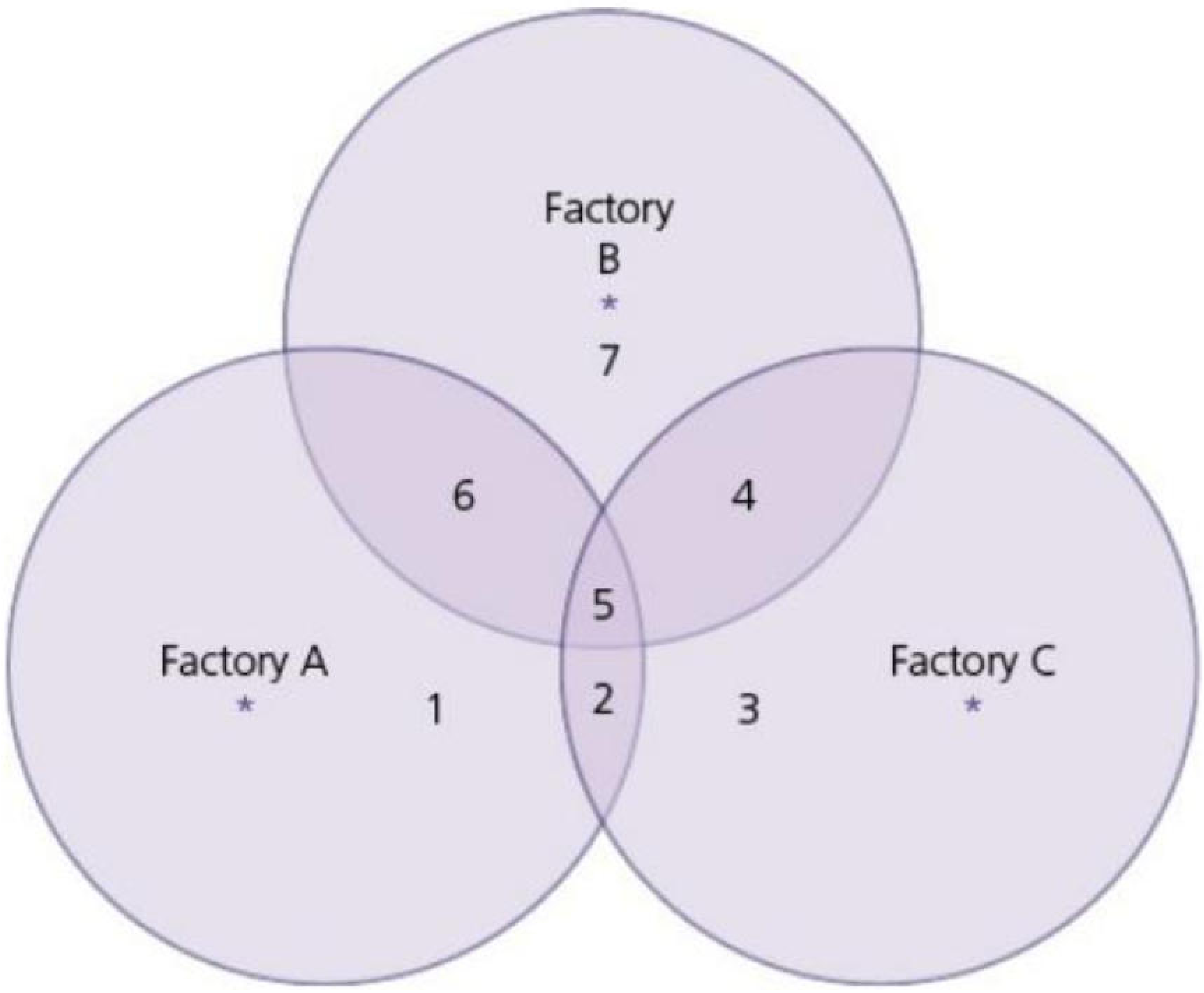
Highly skilled workers, who because of their qualifications have a reasonable choice about where they live and work, will favour areas where the quality of life is high. Influencing factors include:

- proximity to attractive physical environments for recreation
- highly rated schools
- low crime rates
- good quality housing
- a high level of accessibility – roads, railways, airports.

This means that organisations and industries wishing to attract highly skilled workers are likely to take these factors into account when considering where to locate.

Agglomeration economies

Agglomeration is when similar or related industries cluster together in the same location so that ideas and activities can be combined and exchanged. This brings benefits in terms of cost savings, such as reduced transport costs. Figure 9.18 shows that in locations 1, 3 and 7, the three factories are too far away from each other to benefit from any significant cost reductions. In locations 2, 4 and 6, different pairs of factories can benefit from close proximity. However, the area of maximum mutual benefit is location 5, where all three enterprises are in very close proximity.



▲ **Figure 9.18** Agglomeration economies

Agglomeration economies, also known as external economies of scale, can be subdivided into the following:

- » Urbanisation economies: The cost savings resulting from urban location, due to factors such as the range of services available and the investment in infrastructure already in place.
- » Localisation economies: When a firm locates close to its suppliers (backward linkages) or to firms to which it supplies (forward linkages).

Increasingly, manufacturing and large retail companies have located together on industrial estates or **business/retail parks**. These are usually located close to good quality transport infrastructure, often in outer suburban or urban fringe locations.

Technology

Advances in technology have affected all sectors of employment over the last century. For example, large farms and manufacturing industries can now be run by a smaller workforce due to advanced mechanisation and automation. The development of the internet has allowed large companies to outsource production and manage complex operations all over the world. Many jobs in the service industries that were once located in HICs have been outsourced to emerging economies, such as call centres and medical transcription services. Technological advances can also reduce the amount of raw materials required or change the balance between the different raw materials used. Recycling is an important way of reducing the amount of new raw materials used in manufacturing. Such changes can affect location decision-making.

Containerisation

Containerisation has been central to most other transport developments. This system, which has been operating worldwide since the latter part of the last century, involves the use of standardised shipping containers for transport by road, rail and sea. Containerisation has been key to the development of modern intermodal transport terminals. Compared to the traditional way of moving goods from one form of transport to another, the use of containers has substantially reduced costs and the time required for transfers, along with reducing the risk of damage and theft.

A new generation of **deep-sea ports** have been built in recent decades to accommodate the increasing scale of **ultra-large container ships**. Such vessels have a capacity of at least 14,000 TEU (twenty-foot equivalent units). Container ships have increased in size because of the economies of scale that can be achieved. The HMM *Algeciras* class ships (23,964 TEU) are currently the largest container ships in the world. They are 400 m long and 61 m wide.

Activities

1. Which aspects of labour are most important when selecting an area in which to locate?
2. State three factors relating to quality of life that might attract highly skilled labour to a location.
3. Give an example of the way in which technological advance might impact on industrial location.